

AgroSphere – Detailed Answers (Module 16 Assessment)

1. System Architecture (MVT in Django)

Django follows the MVT (Model-View-Template) architecture. Models handle database structure such as User, Product, and Booking. Views contain business logic like authentication, booking, and payments. Templates handle UI using HTML, CSS, and Bootstrap.

2. User Management Implementation

User authentication can be implemented using Django's built-in authentication system. Features include signup, login, logout, password reset via email/SMS, and profile management with document upload using Django FileField.

3. Product & Resource Catalog

Products like seeds, tools, and fertilizers are stored in models with fields such as name, category, price, and description. Filtering can be implemented using Django ORM queries. Guides can include videos and PDFs uploaded via media files.

4. Expert Booking System

Farmers can book consultations using a Booking model. It includes date, time, expert, and status. Availability checking is done before confirmation. Booking history is stored for future reference.

5. Admin Panel Features

Django Admin allows managing users, products, and bookings. Custom roles like Admin, Agronomist, and Sales Support can be implemented using Groups and Permissions. Reports can be generated using charts.

6. Payment Integration

Payment gateways like Razorpay or Paytm can be integrated using APIs. After successful payment, order status is updated in the database.

7. Deployment Readiness

Project can be deployed using platforms like PythonAnywhere or Render. Use virtualenv, .env files for security, and GitHub for version control. Database can be SQLite or PostgreSQL.

8. Regional Language Support

Django supports internationalization (i18n). You can translate content into multiple languages to help farmers understand better.

9. Weather API Integration

Weather APIs like OpenWeather can be integrated to show forecasts based on crop zones, helping farmers plan activities.

10. Post-Consultation Features

Experts can upload prescriptions and documents using file upload features. These can be accessed by farmers later from their dashboard.

11. Mobile App & Offline Support

The platform can scale to mobile apps using frameworks like React Native or Flutter. Offline support can be added using local storage and sync mechanisms.